

INSTAGRAM DATASET

Exploration & Analytics Report

Social Media Analytics Framework • SIH Prototype

29,999

records

23

columns / signals

0

missing values

Prepared from the Instagram CSV schema and analysis shared.

Executive Summary

What this dataset gives the prototype

29,999

Instagram post records

23

available data fields

13

numeric fields

9

text/category fields

Core observation

- The dataset is already structurally complete: every listed column has 29,999 non-null values.
- It contains content, account, timing, traffic, engagement and performance signals.
- This is strong enough for exploratory analysis, feature engineering and a prototype ML pipeline.

Why it matters for SIH

- Instagram-specific strengths: reach, impressions, saves, shares and followers gained.
- These signals let us study not only likes, but deeper content performance and audience response.
- The next layer is NLP + trend/anomaly analysis + performance prediction.

Dataset Overview

Schema-level analysis from the supplied Pandas output

Identity

post_id • account_id • account_type

Audience

follower_count

Content

media_type • content_category • caption_length • hashtags_count

Distribution

traffic_source • has_call_to_action

Time

post_datetime • post_date • post_hour • day_of_week

Engagement

likes • comments • shares • saves

Exposure

reach • impressions

Growth

followers_gained

Outcome

engagement_rate • performance_bucket_label

Data Quality & Preparation

The supplied DataFrame is clean at the structural level

29,999 / 29,999

non-null records across each shown column

23

total columns

5.3 MB

reported DataFrame memory

Recommended validation

- No null values are reported in the provided DataFrame.info() output.
- Next validation: duplicate post_id checks, range checks and datatype conversion.
- post_datetime and post_date are currently strings and should be parsed to datetime.

Quality checks before ML

- Check follower_count, reach, impressions and engagement values for impossible negatives.
- Verify engagement_rate calculation against its source metrics.
- Inspect category spelling/casing for account_type, media_type, traffic_source and labels.

Pandas Operations Performed / Recommended

From raw CSV → analysis-ready table

1

Inspect

head(), shape, columns, info(), describe()

2

Validate

isnull(), duplicated(), unique(), value_counts()

3

Clean

drop_duplicates(), fillna(), type conversion

4

Transform

datetime extraction, normalization, derived metrics

5

Group

groupby() for account, media, category, time

6

Rank

sort_values() for top posts/accounts/categories

7

Feature

engagement, ratios, text length, hashtag signals

8

Export

analysis table → CSV/Parquet → ML pipeline

Content & Media Analysis

Understand what type of content performs best

Media + Category

- Media type distribution
- Content category distribution
- Average likes/comments/shares/saves by media type
- Engagement rate by content category
- Reach and impressions by content category
- Followers gained by content type

Caption + Hashtags

- Caption length distribution
- Hashtag count distribution
- Compare short vs long captions
- Compare low vs high hashtag usage
- Identify combinations associated with strong performance

Top Performers

- Top posts by engagement rate
- Top posts by reach
- Top posts by shares
- Top posts by saves
- Top posts by follower growth
- Detect unusually high-performing posts

Engagement & Reach Analysis

Move beyond likes: measure meaningful audience response

Likes

attention / reaction

Comments

conversation

Shares

propagation

Saves

content value

Reach

unique exposure

Impressions

total exposure

Key derived insight

- Recommended derived metric: $\text{engagement_rate} = (\text{likes} + \text{comments} + \text{shares} + \text{saves}) / \text{reach} \times 100$ — use the dataset's existing definition if it differs.

Audience & Account Analysis

Who is posting, and how audience size relates to performance

Account-level analysis

- Follower count distribution
- Account type distribution
- Average engagement by account type
- Follower count vs reach
- Follower count vs engagement rate
- Followers gained per post / per reach

Useful rankings

- Top accounts by average engagement
- Top accounts by total reach
- Top accounts by follower growth
- Posting frequency per account
- Identify high-efficiency accounts rather than only large accounts

Timeline & Best-Time Analysis

Discover when the audience is most active

post_datetime

parse timestamp

post_date

daily volume

post_hour

hourly activity

day_of_week

weekday pattern

Questions to answer

- Group posts by hour/day and calculate average engagement_rate.
- Identify peak posting windows and peak performance windows separately.
- Peak activity $\neq$ peak performance — compare both.

Trend / anomaly layer

- Detect unusual daily/hourly spikes.
- Compare media types across time windows.
- Use rolling averages or anomaly detection for future live-data ingestion.

Traffic Source & Call-to-Action Analysis

Which distribution channels and CTAs contribute to performance?

Traffic source

- traffic_source frequency
- Average reach by traffic source
- Average engagement rate by source
- Shares and saves by source
- Followers gained by source
- Find the most efficient acquisition/distribution source

CTA effectiveness

- Compare posts with and without call-to-action.
- Measure differences in comments, shares, saves and followers gained.
- Check whether CTA impact changes by media_type or content_category.
- Avoid claiming causation from observational data alone.

NLP: Sentiment & Trend Analysis

The supplied schema does not include raw caption text — add text data for this module

Current capability

- Current fields: caption_length and hashtags_count.
- These support structural content analysis.
- Raw caption text is not listed among the 23 columns supplied.

Sentiment pipeline

- If raw captions are available: clean text, tokenize, remove noise.
- Run sentiment classification: positive / neutral / negative.
- Store sentiment label + confidence score.

Trend pipeline

- Extract hashtags and keywords.
- Track topic frequency over time.
- Detect rising topics / sudden narrative changes.
- Combine topic + sentiment + engagement.

Performance Prediction & ML Readiness

Use the dataset to predict or classify post performance

Target variable

- Target: performance_bucket_label
- Possible classes: inspect actual unique labels before training.
- This is a supervised classification setup if the label is derived before modeling.

Candidate features

- followers, media type, category
- traffic source, CTA
- post hour/day
- caption length, hashtags
- Reach/impressions only if they are available at prediction time

Model layer

- Baseline: Logistic Regression / Decision Tree
- Compare with Random Forest / Gradient Boosting
- Train/test split
- Accuracy + precision + recall + F1
- Feature importance / explainability

Important ML Design Decision

Prevent data leakage before claiming a predictive model

Leakage risk

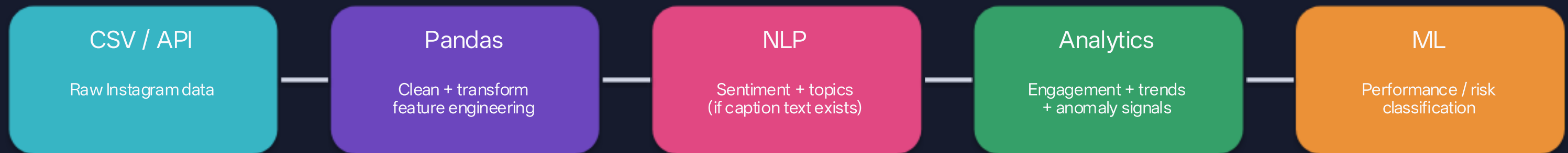
- If performance_bucket_label is created from engagement_rate, do not use engagement_rate as an input feature.
- Likewise, likes/comments/shares/saves may be post-outcome variables and can leak the answer.
- Decide what information is known at prediction time: before posting, shortly after posting, or after the campaign.

Correct framing

- For pre-post prediction: use account, content category, media type, traffic source, CTA, timing, caption/hashtags.
- For post-publication performance classification: engagement variables can be used, but the use-case changes.
- Document the prediction timestamp in the final system.

How This Fits the SIH Social Media Analytics Framework

Instagram prototype → modular analytics pipeline



Prototype outputs

- Dashboard: account performance, content performance, sentiment, topics and trends.
- Alerts: unusual activity, sudden negative-sentiment shifts, rapidly rising narratives and anomalous engagement.
- Network/propagation extension: use shares and account relationships where relationship data is available.

Limitations & Next Data Requirements

What this CSV can and cannot demonstrate

Current limitations

- The provided schema has no raw caption text, so direct sentiment/topic NLP cannot be demonstrated from these 23 fields alone.
- No explicit follower-to-follower relationship field is present, so full network analysis is limited.
- Demographic inference (age, gender, location, profession) is not supported by this schema.
- Historical CSV analysis is different from a live monitoring system.

Next useful data

- Raw caption text / hashtags, if legally and ethically available.
- Account relationship / interaction data for propagation graphs.
- Timestamped continuous ingestion for real-time trend detection.
- Ground-truth labels for threat/risk classification if that becomes a project objective.

Final Takeaway

From 29,999 Instagram records to an explainable analytics system

DATA → FEATURES → INSIGHTS → ML → DASHBOARD

1 • Understand

- Clean & validate
- Engineer meaningful features
- Compare content / audience / time

2 • Analyze

- Find high-performing content
- Detect trends & anomalies
- Add NLP when text is available

3 • Build

- Predict performance
- Explain important features
- Expose insights in dashboard

This deck intentionally does not invent numerical findings that were not included in the shared analysis output.